ECHEM Assignment 1: Cycling a Commercial $LiMn_2O_4$ Cathode

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Experimental details

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A commercial $LiMn_2O_4$ (LMO) cathode was cycled by cyclic voltammetry at three scan rates and by galvanostatic charge/discharge at four C-rates. All seven tests were run on the same cell, in the order listed below. All potentials are reported vs. Li/Li^+.

Item	Detail
Electrode	Commercial LMO sheet (MTI Corp.): 15 µm Al foil, 85 µm single-side coating, 94.2 wt% active material, 110 mAh/g rated
Cell capacity used for C-rates	1.434 mAh (EC-Lab cell setting)
Instrument	Bio-Logic VMP-300, EC-Lab V11.33, Channel 10, operated remotely via RemotePC
CV settings	Start 3.000 V, vertices 4.500 V and 3.000 V (4.800 V at 0.3 mV/s), 2 cycles, I range Auto
Galvanostatic settings	Charge to 4.5 V, discharge to 3.5 V, 30 s rest after each step, record every 10 mV or 5 s, 2 cycles
Data handling	Exported from EC-Lab as text; plotted on a personal computer

#	Date started	Test	Current or scan rate	Window (V)	Duration	Data file
1	Sep 28, 2026	Galvanostatic, C/10	0.143 mA	3.5–4.5	41.3 h	gal test 2_C10
2	Sep 30, 2026	Galvanostatic, C/5	0.287 mA	3.5–4.5	20.0 h	Soham_GCD_C5_C10

#	Date started	Test	Current or scan rate	Window (V)	Duration	Data file
3	Oct 1, 2026	Galvanostatic, C/2	0.717 mA	3.5–4.5	7.7 h	Soham_GCD_C2_C10
4	Oct 1, 2026	CV	0.2 mV/s	3.0–4.5	8.3 h	Soham_CV_0.2_C10
5	Oct 2, 2026	CV	0.3 mV/s	3.0–4.8	6.7 h	Soham_CV_0.3_C10
6	Oct 2, 2026	Galvanostatic, 2C	2.868 mA	3.5–4.5	1.3 h	Soham_GCD_2C_C10
7	Oct 3, 2026	CV	0.1 mV/s	3.0–4.5	14.0 h	Soham_CV_0.1_C10

Question 1 — Cyclic voltammetry

As the scan rate increases, the oxidation peaks shift to higher potential and the reduction peaks shift to lower potential. The separation between them widens from 0.55 V at 0.1 mV/s to 0.86 V at 0.3 mV/s.

Q1a–c: Voltage and current vs. time

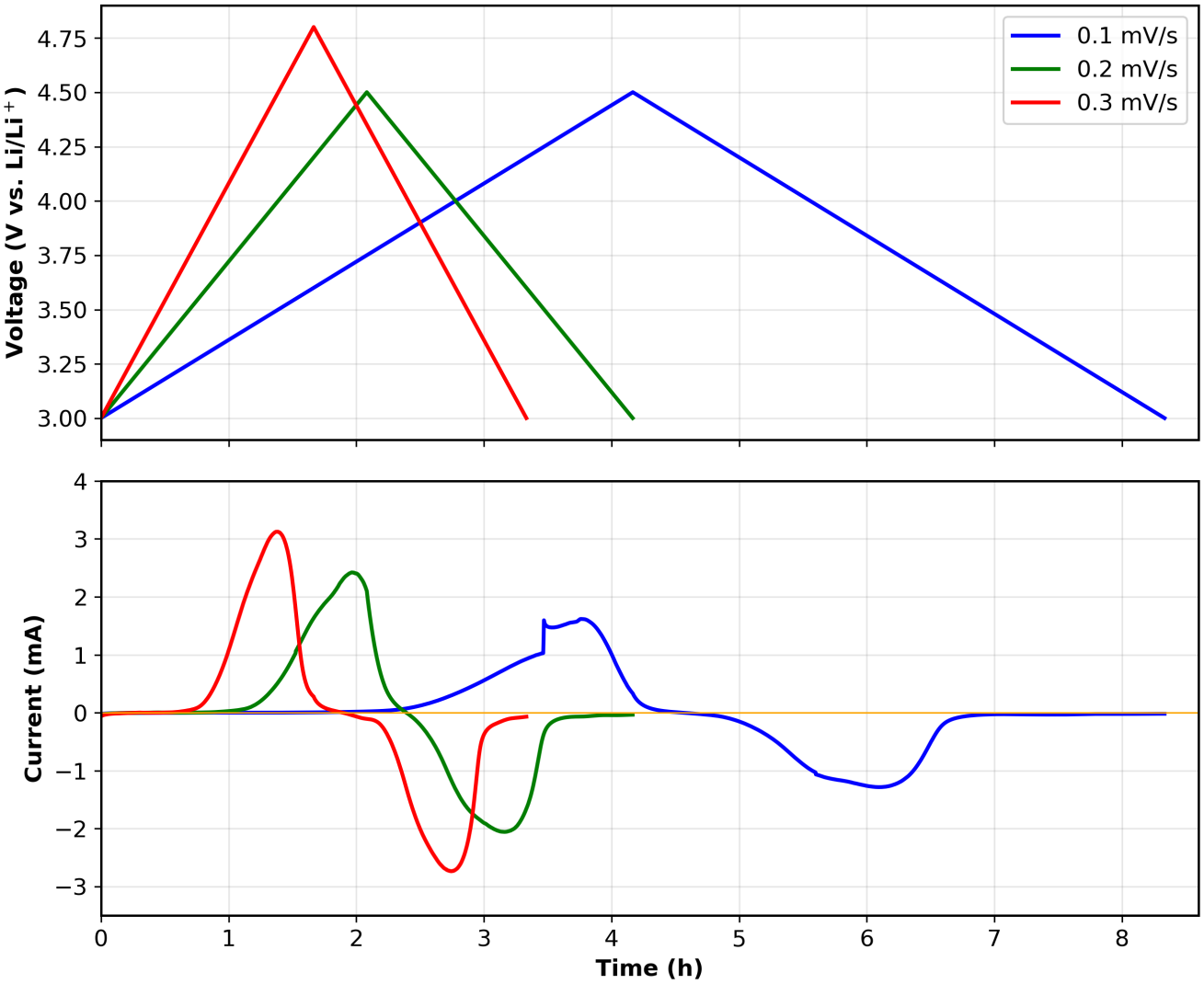


Figure 1. One full CV cycle at each scan rate (3.0 V → vertex → 3.0 V), with time measured from the start of that cycle. Upper: voltage vs. time. Lower: current vs. time. Blue 0.1 mV/s, green 0.2 mV/s, red 0.3 mV/s.

The voltage ramps are linear at the set scan rates, so one cycle takes 8.3 h, 4.2 h and 3.3 h (the 0.3 mV/s cycle extends to 4.8 V). Each anodic sweep produces a positive current peak, from Li^+ extraction ($LiMn_2O_4 \rightarrow Li_{1-x}Mn_2O_4 + xLi^+ + xe^-$). Each cathodic sweep produces a negative peak, from Li^+ reinsertion. Faster scans give taller peaks in less time.

Q1d: Current vs. voltage

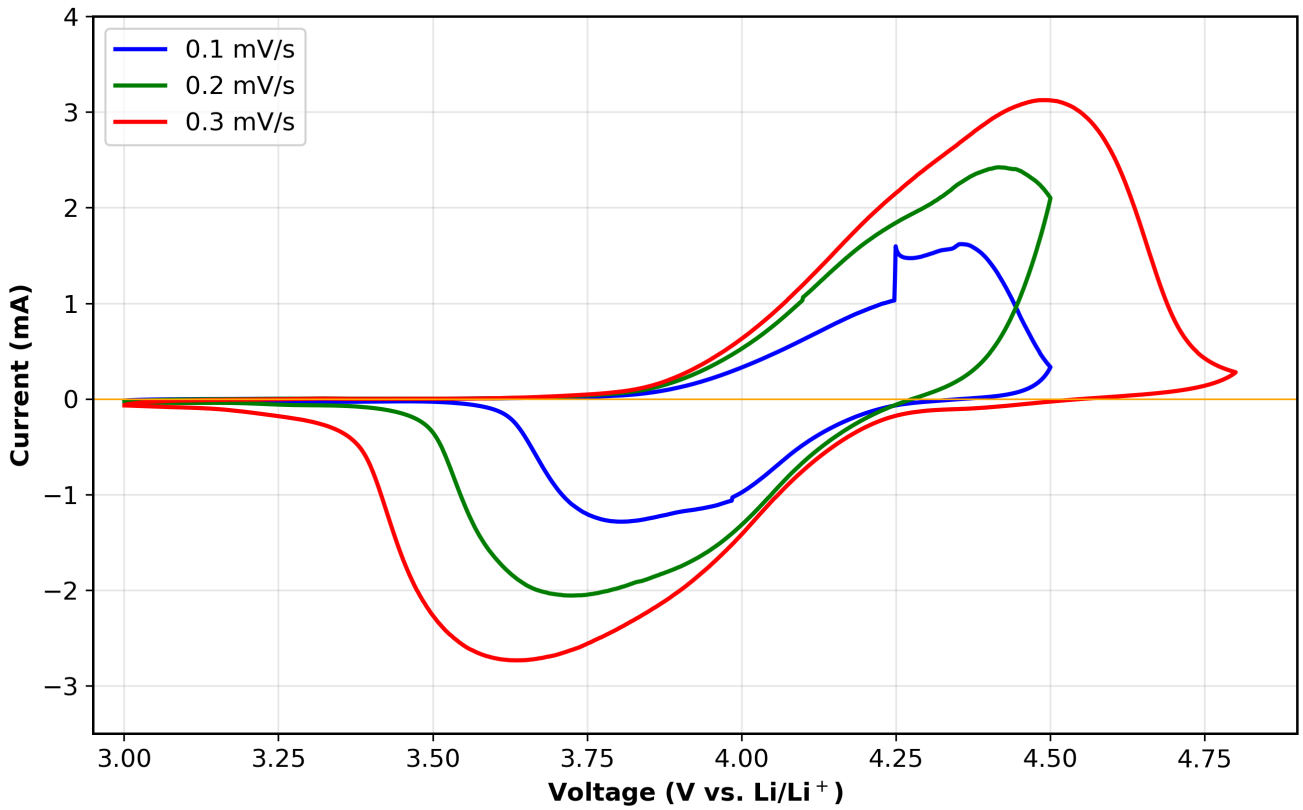


Figure 2. Current vs. voltage for the same three cycles as Figure 1. Blue 0.1 mV/s, green 0.2 mV/s, red 0.3 mV/s.

Scan rate (mV/s)	Oxidation peak (V)	Oxidation peak current (mA)	Reduction peak (V)	Reduction peak current (mA)	Peak separation (V)
0.1	4.36	1.62	3.81	−1.28	0.55
0.2	4.42	2.42	3.72	−2.06	0.69
0.3	4.49	3.12	3.64	−2.73	0.86

In what direction do the oxidation peaks shift with increasing scan rate? To higher (more positive) potential: 4.36 → 4.42 → 4.49 V.

In what direction do the reduction peaks shift with increasing scan rate? To lower (less positive) potential: 3.81 → 3.72 → 3.64 V.

At a faster scan rate, there is less time at each potential for charge transfer at the LMO/electrolyte interface and for Li^+ diffusion through the LMO particles. A larger overpotential is therefore needed to drive the reaction. The ohmic (IR) drop also grows,

because the peak currents are larger (1.62 → 3.12 mA). Both effects push the anodic peak up and the cathodic peak down, so the redox process becomes less reversible as the scan rate increases.

Question 2 — Galvanostatic charge–discharge

As the C-rate increases, both the capacity and the energy delivered by the cell decrease. At 2C the cell delivers 58% of its C/10 capacity and 53% of its C/10 energy.

Q2a: C/10

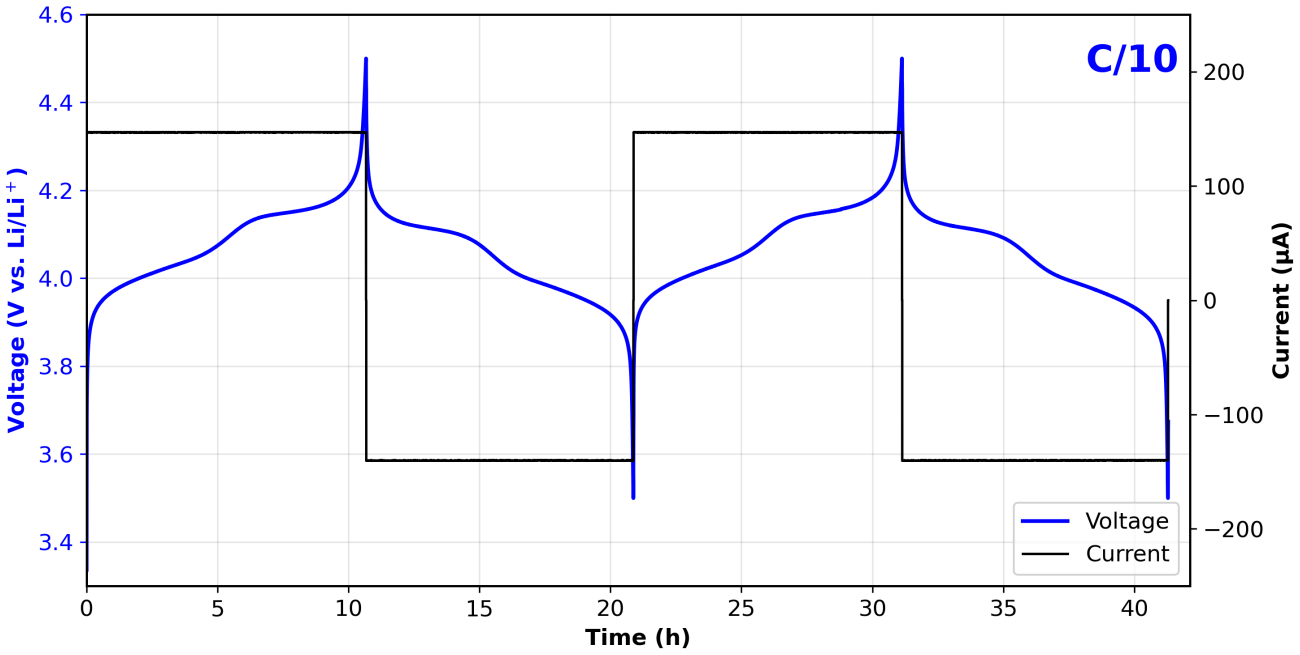


Figure 3. C/10 (±0.143 mA): voltage vs. time (blue) and current vs. time (black), two cycles over 41.3 h.

Each half-cycle takes about 10 h. The two voltage plateaus near 4.0 V and 4.15 V are characteristic of spinel LMO, which releases and takes up lithium in two stages.

Q2b: C/5

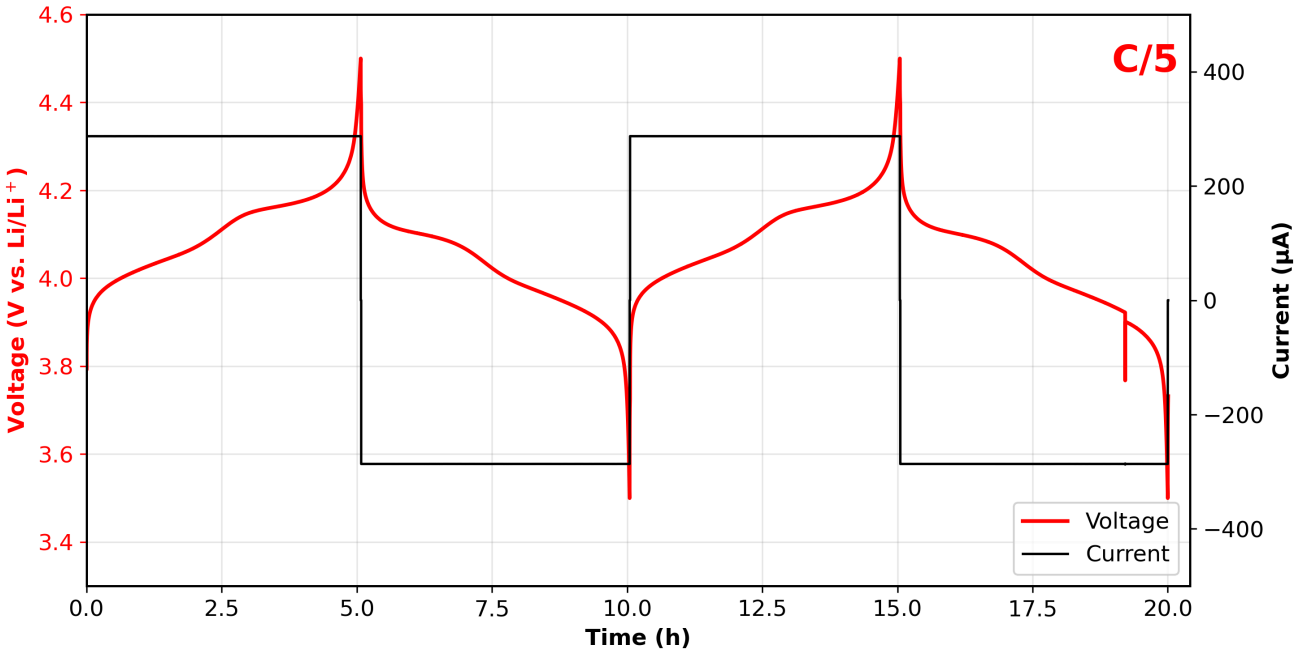


Figure 4. C/5 (±0.287 mA): voltage vs. time (red) and current vs. time (black), two cycles over 20.0 h.

Q2c: C/2

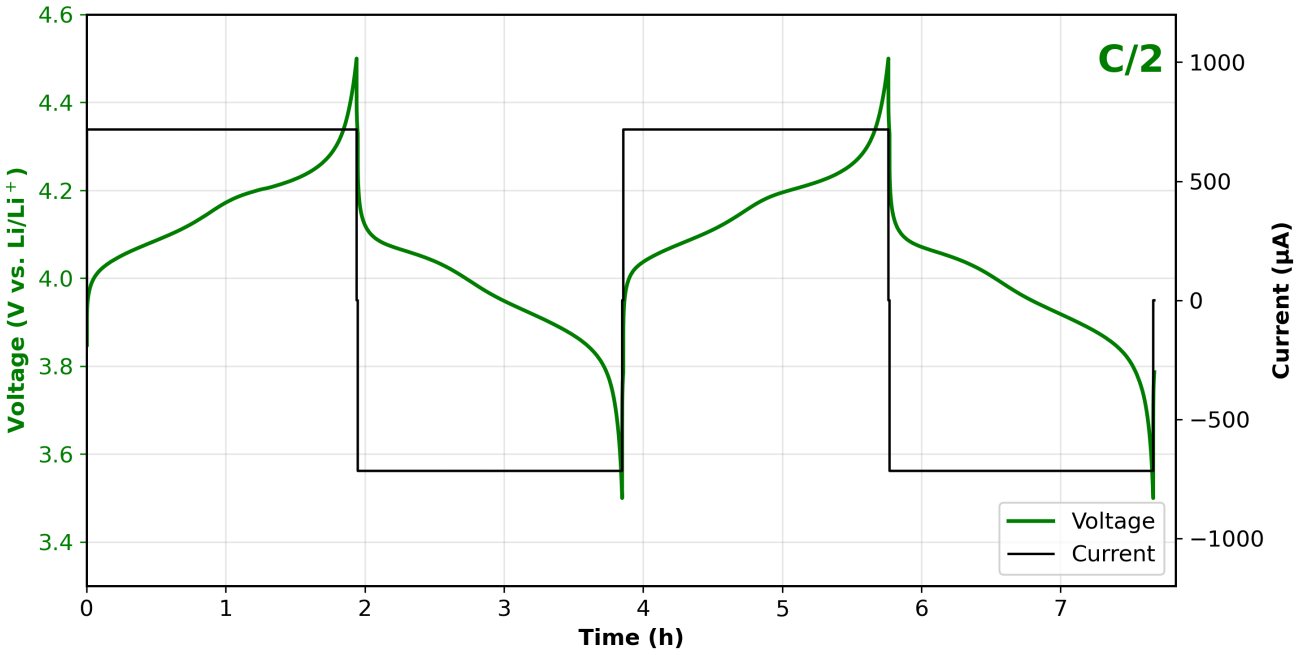


Figure 5. C/2 (±0.717 mA): voltage vs. time (green) and current vs. time (black), two cycles over 7.7 h.

Q2d: 2C

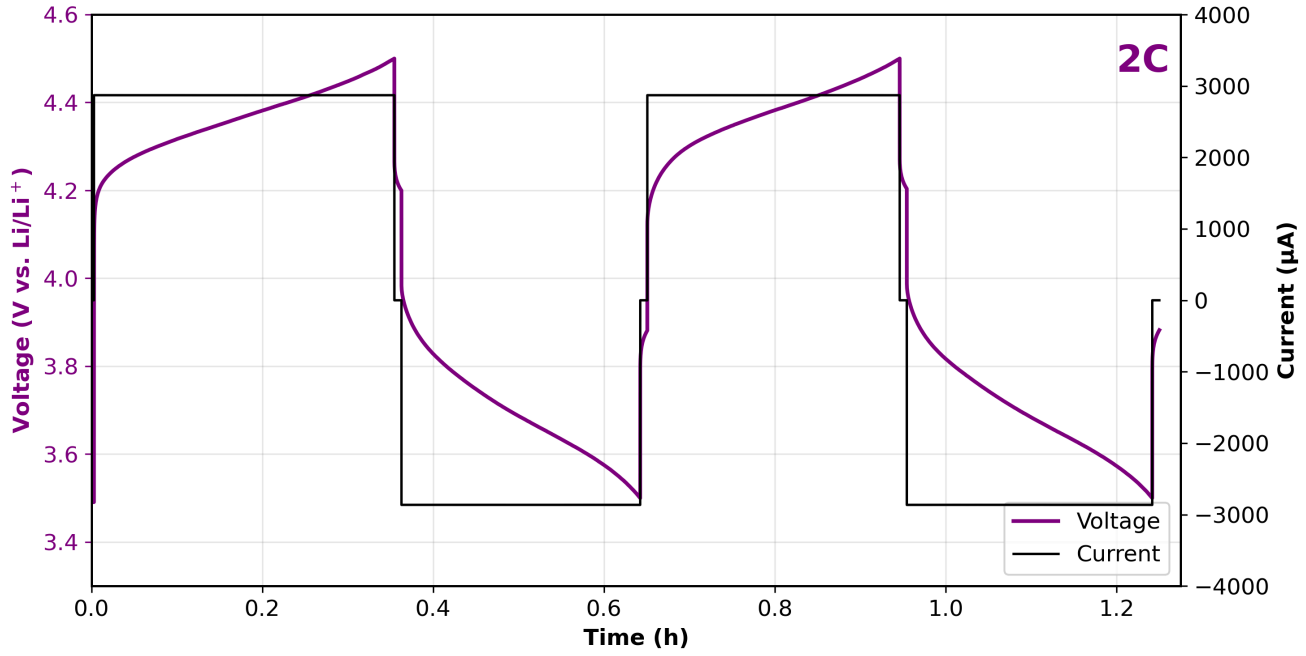


Figure 6. 2C (±2.868 mA): voltage vs. time (purple) and current vs. time (black), two cycles over 1.3 h.

At 2C the two plateaus are no longer distinct. The voltage jumps sharply when the current switches on or off, a sign of large ohmic and kinetic losses at this current.

Q2e: Voltage vs. time, all C-rates

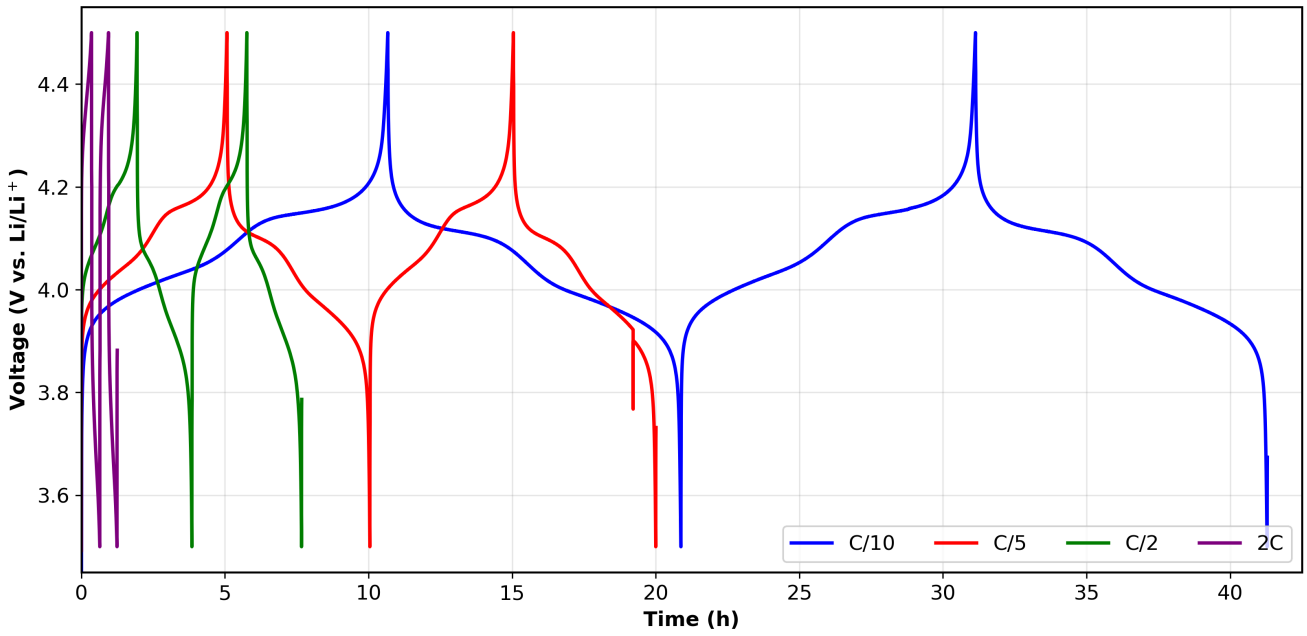


Figure 7. Voltage vs. time for all four C-rates. Blue C/10, red C/5, green C/2, purple 2C.

Q2f: Voltage vs. capacity, all C-rates

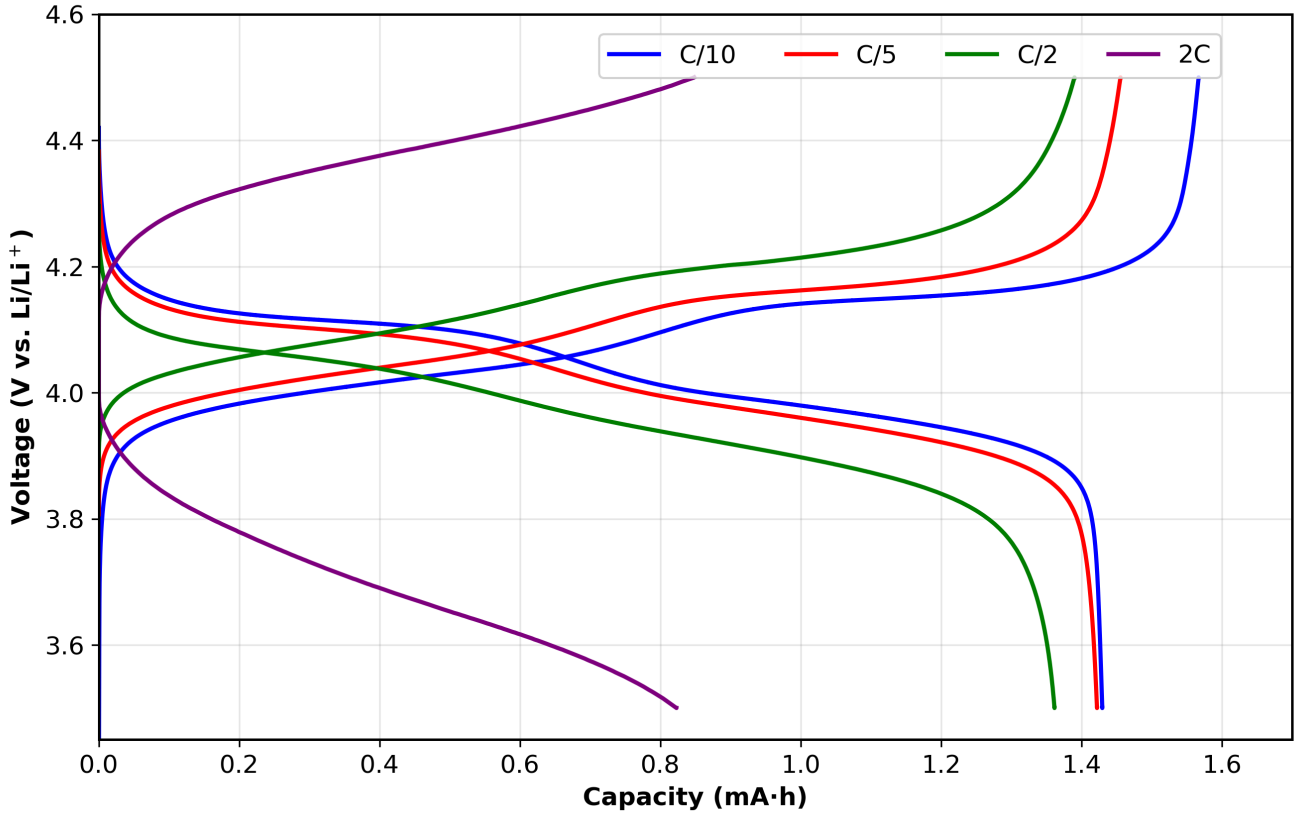


Figure 8. Voltage vs. capacity for one charge–discharge cycle at each C-rate. Blue C/10, red C/5, green C/2, purple 2C.

C-rate	Current (mA)	Cycle plotted	Charge capacity (mAh)	Discharge capacity (mAh)	Avg. charge voltage (V)	Avg. discharge voltage (V)	Discharge energy (mWh)	Discharge energy vs. C/10
C/10	0.143	1	1.567	1.430	4.09	4.04	5.77	100%
C/5	0.287	1	1.455	1.422	4.11	4.02	5.71	99%
C/2	0.717	1	1.390	1.361	4.16	3.96	5.39	93%
2C	2.868	2	0.848	0.823	4.37	3.70	3.04	53%

What happens to the capacity when the charge–discharge rate increases? It decreases. Discharge capacity falls from 1.43 mAh at C/10 to 1.42, 1.36 and 0.82 mAh at C/5, C/2 and 2C. The loss is small up to C/2, then sharp: 2C delivers only 58% of the C/10 capacity. The

overpotentials (ohmic, charge-transfer and Li^+ diffusion through the LMO particles and electrolyte) grow with current. The cell therefore reaches the 4.5 V charge cutoff and the 3.5 V discharge cutoff before lithium extraction or insertion is complete, leaving capacity unused.

What happens to the energy of a battery when it is charged or discharged very quickly? It decreases, and more steeply than the capacity. Since $E = q \cdot V$, both factors fall together. The discharge capacity drops from 1.43 to 0.82 mAh, and the average discharge voltage drops from 4.04 to 3.70 V. Discharge energy therefore falls from 5.77 mWh at C/10 to 3.04 mWh at 2C, 53% of the slow-rate value.

Fast charging also costs more energy, because the average charge voltage rises from 4.09 V to 4.37 V. The gap between charge and discharge voltage widens from 0.05 V at C/10 to 0.68 V at 2C. Round-trip energy efficiency falls from 95% at C/5 to 82% at 2C. The difference is dissipated as heat.

Data notes

Five features of the raw data affect how the figures were prepared. None changes the answers above.

- **Cycle plotted.** One cycle per rate is shown, as the assignment permits. For CV, this is the full 3.0 V → vertex → 3.0 V cycle (cycle 2). For galvanostatic tests, it is cycle 1, except at 2C. The 2C test followed a CV that ended at 3.0 V, so its first charge removed extra lithium (1.01 vs. 0.85 mAh); cycle 2 is the representative 2C cycle.
- **0.1 mV/s CV start.** This run began from the rest potential (3.96 V) rather than 3.0 V. It still contains one complete 3.0 → 4.5 → 3.0 V cycle, which is the cycle plotted in Figures 1 and 2.
- **Current-range switching.** The CV current range was set to Auto, so small steps appear near ±1 mA when the instrument changes range. At 0.1 mV/s, the anodic current held near 1.03 mA from about 4.20 to 4.25 V, then jumped to 1.6 mA at 4.25 V. This is an instrument artifact, not a redox peak; the oxidation peak at 4.36 V lies beyond it.
- **Contact disturbance at C/5.** At 19.2 h, during the second C/5 discharge, the voltage fluctuated for about 1 s and settled about 22 mV lower for the rest of that discharge, with the current unchanged. It is visible in Figures 4 and 7. The cell resistance was back to normal in the following C/2 test, and Figure 8 uses C/5 cycle 1, recorded before the event.

- **C/10 current measurement.** The C/10 test used the 10 mA current range, which reads small currents less precisely: +0.147 mA on charge and −0.140 mA on discharge vs. 0.143 mA set. Its first-cycle charge capacity and coulombic efficiency (91%) are therefore approximate. The C/5 and C/2 tests used the 1 mA range and held the set current within 0.3%.

Conclusions

The commercial LMO cathode shows the expected rate dependence in both techniques. In CV, the oxidation peak moves from 4.36 to 4.49 V and the reduction peak from 3.81 to 3.64 V as the scan rate triples, widening the peak separation from 0.55 to 0.86 V. In galvanostatic cycling, raising the rate from C/10 to 2C cuts the discharge capacity to 58% and the discharge energy to 53%, because the growing overpotentials lower the discharge voltage and stop each step at its cutoff early.

Both results trace back to the same cause: at higher current, ohmic, charge-transfer and diffusion losses grow, so less of the stored lithium is used and less of the stored energy is recovered.